

Business Problem Document

Executive Summary

Employee attrition has a direct impact on organizational performance by increasing recruitment costs, training expenses, workload on existing employees, and the loss of experienced talent. This project develops an end-to-end HR analytics solution using Python, SQL, and Power BI to transform raw employee data into meaningful business insights. The interactive dashboard enables HR managers to monitor workforce metrics and identify patterns that support strategic decision-making.

Business Problem

Organizations often struggle to understand why employees leave and which workforce segments are most affected. Without centralized reporting and analysis, HR teams cannot quickly identify departments, job roles, age groups, or employee categories with high attrition. This limits their ability to implement targeted retention strategies and optimize workforce planning.

Business Challenges

- High employee turnover increases recruitment and onboarding costs.
- Loss of experienced employees reduces productivity and business continuity.
- Difficulty identifying high-risk departments and job roles.
- Limited visibility into workforce trends and employee demographics.
- Need for interactive reporting to support HR decision-making.

Business Objectives

- Measure employee attrition using key HR KPIs.
- Analyse attrition by department, job role, gender, age group, and years at the company.
- Provide interactive dashboards for workforce exploration.
- Support evidence-based HR planning and employee retention initiatives.

Project Scope

The project covers the complete analytics workflow, including data preparation in Python, analytical querying in SQL, and interactive dashboard development in Power BI using DAX measures and visualizations.

Project Deliverables

- Cleaned HR dataset
- SQL analysis queries

- Interactive Power BI dashboard
- Business problem document
- HR Analytics Analysis Report

Expected Business Value

The solution helps HR professionals identify attrition patterns, monitor workforce KPIs, prioritize retention initiatives, and support strategic decisions through data-driven insights.